

- Sanity check: signs, units, conversions, mappings
- Mass balance and convergence checks

DIAGNOSTICS & VALIDATION ITERATIVE IN (1) – (5)

- Investigate outliers and ensure predictions match data
- Identify uncertainty hotspots, drivers → prioritize new data

1. SYSTEM DEFINITION

- Boundaries (space / time / scope)
- Define processes, flows, and units
- Graph $G = (P, F)$; set balance constraints

2. DATA ACQUISITION & QUALITY

- Compile datasets from statistics, reports
- Convert units / basis / time frames
- Assign data sources and pedigree scores → uncertainty ranges

3. PRIORS & LIKELIHOODS

- Specify positivity and share constraints
- Select distributions (Dirichlet, log-normal, normal)
- Map pedigree scores → covariance Σ

4. INFERENCE

- MAP (QP) or MCMC (HMC / NUTS)
- Impose mass-balance $Ax = b$, $x \geq 0$

5. UNCERTAINTY- AWARE REPORTING

- Posterior means ± credible intervals
- Sankey diagrams, flow maps, KPIs (e.g., recycling rate and recycled content)
- Scenario analysis

INTEGRATION WITH OTHER ANALYSIS

- LCA for consumption-based impacts with uncertainty propagation
- Environmental IO for production-based impacts with uncertainty propagation

- Spatial analysis of scrap generation and recycling (capacity expansion planning)
- Technology systems analysis for improving key circularity indicators and embodied impacts